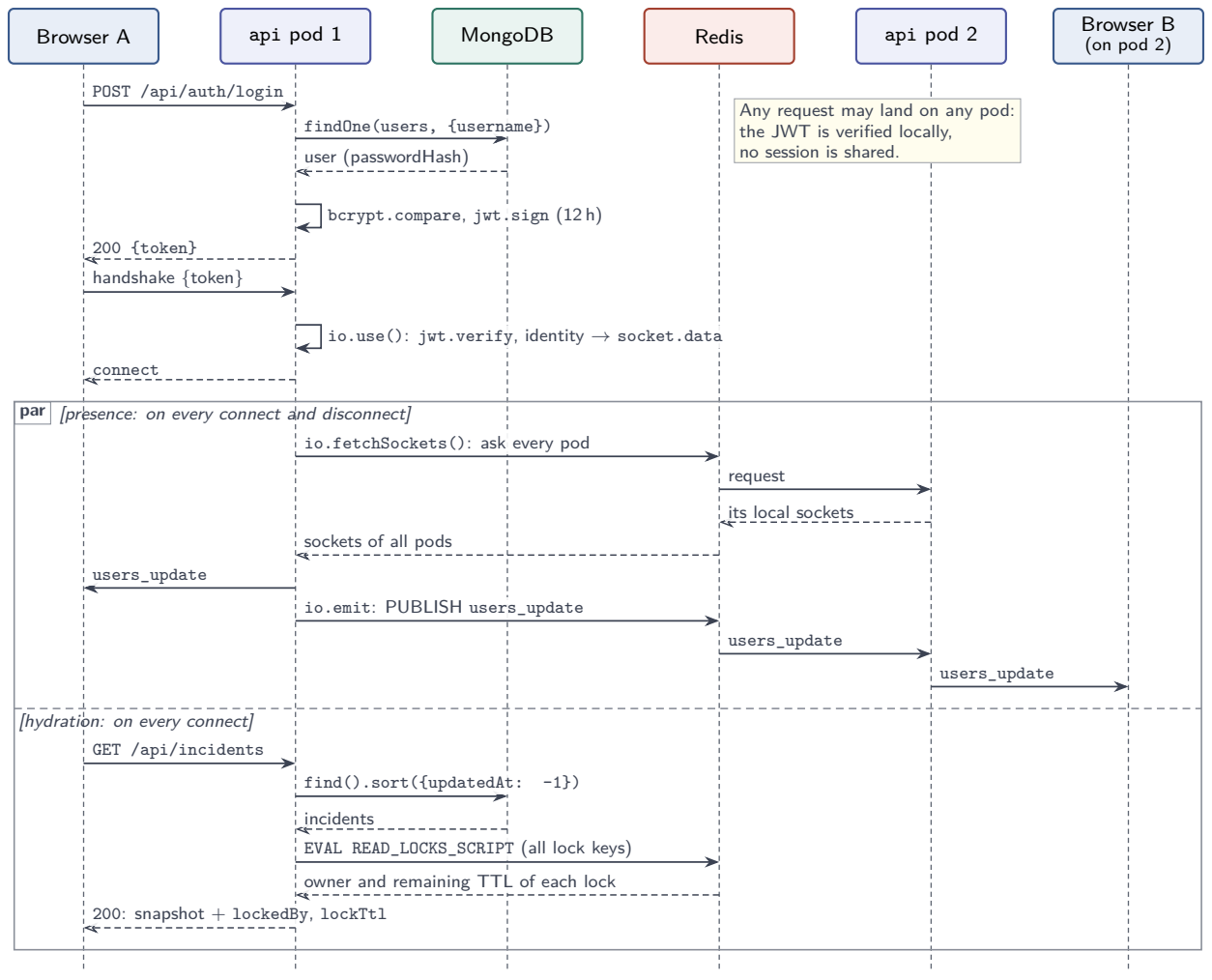
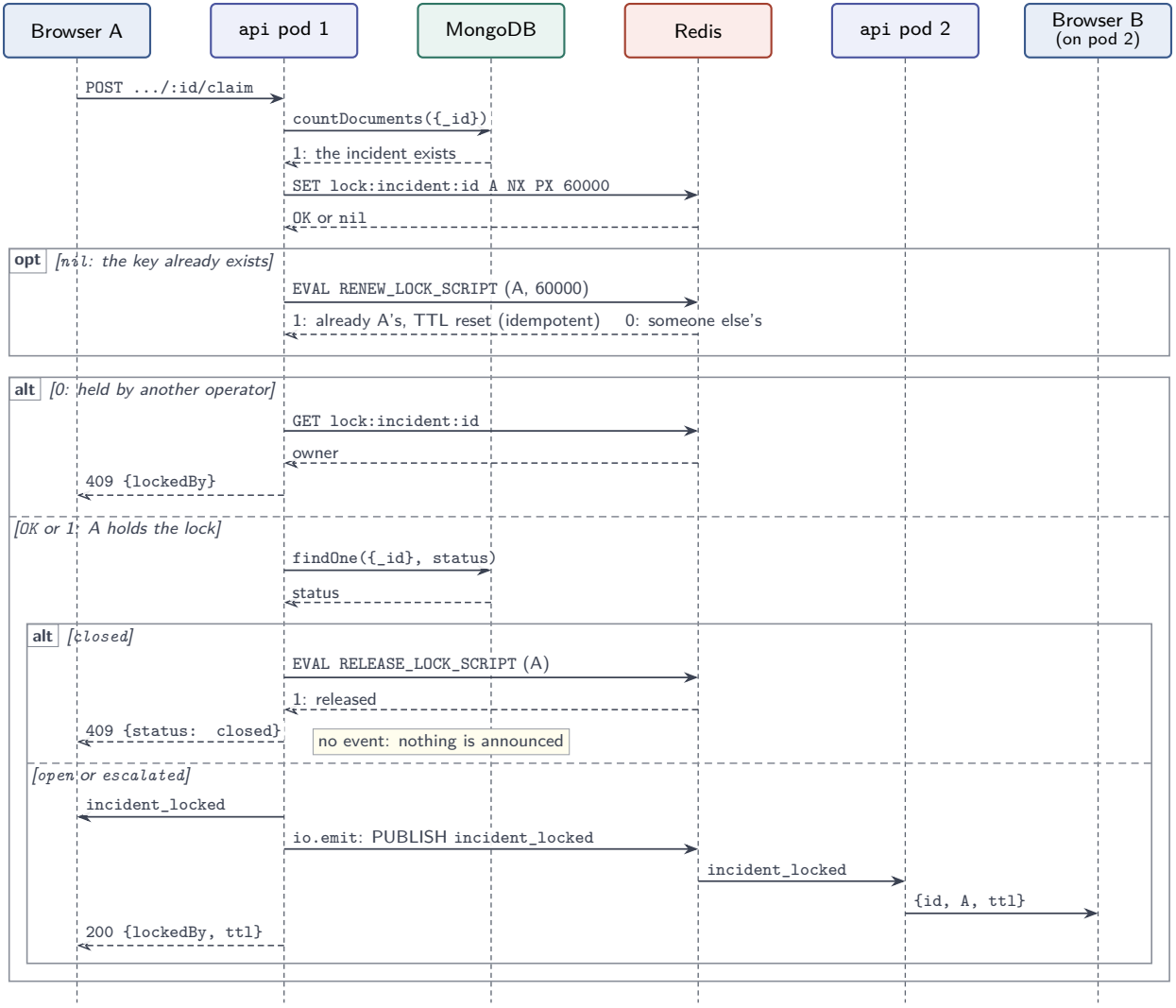
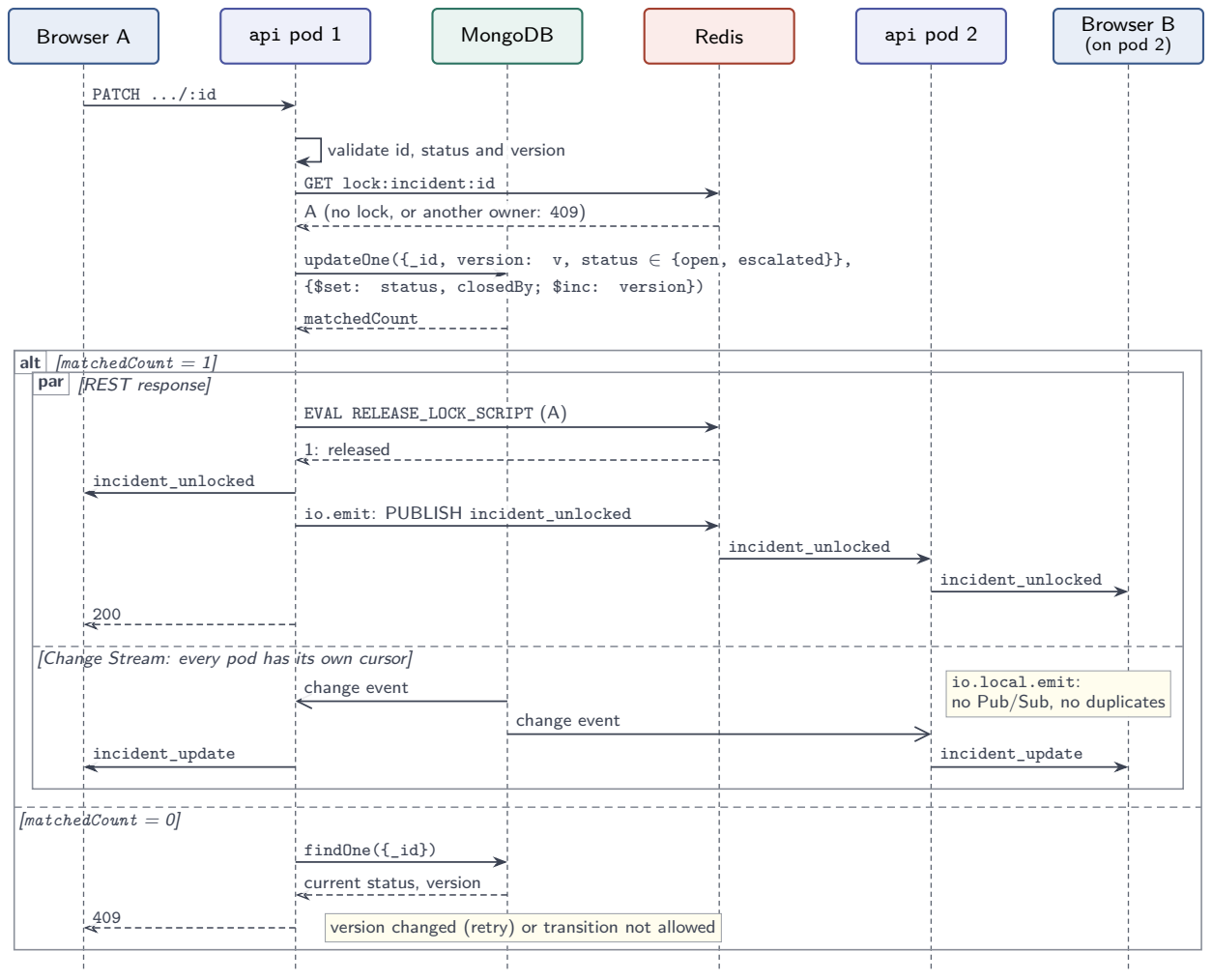
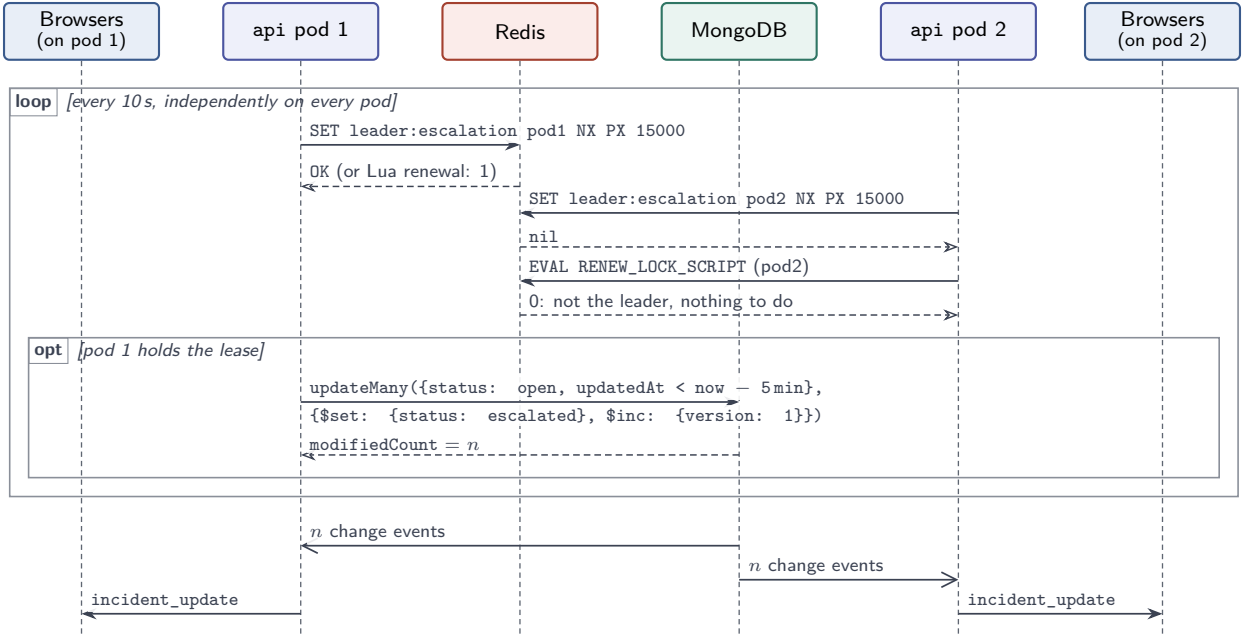

Browser A
api pod 1
MongoDB
Redis
api pod 2
Browser B
(on pod 2)
POST /api/auth/login
findOne(users, {username})
user (passwordHash)
Any request may land on any pod:
the JWT is verified locally,
no session is shared.
bcrypt.compare, jwt.sign (12 h)
200 {token}
handshake {token}
io.use(): jwt.verify, identity → socket.data
connect
par
[presence: on every connect and disconnect]
io.fetchSockets(): ask every pod
request
its local sockets
sockets of all pods
users_update
io.emit: PUBLISH users_update
users_update
users_update
[hydration: on every connect]
GET /api/incidents
find().sort({updatedAt: -1})
incidents
EVAL READ_LOCKS_SCRIPT (all lock keys)
owner and remaining TTL of each lock
200: snapshot + lockedBy, lockTtl


Browser A
api pod 1
MongoDB
Redis
api pod 2
Browser B
(on pod 2)
POST .../:id/claim
countDocuments({_id})
1: the incident exists
SET lock:incident:id A NX PX 60000
OK or nil
opt
[nil: the key already exists]
EVAL RENEW_LOCK_SCRIPT (A, 60000)
1: already A's, TTL reset (idempotent)
0: someone else's
alt
[0: held by another operator]
GET lock:incident:id
owner
409 {lockedBy}
[OK or 1: A holds the lock]
findOne({_id}, status)
status
alt
[closed]
EVAL RELEASE_LOCK_SCRIPT (A)
1: released
409 {status: closed}
no event: nothing is announced
[open or escalated]
incident_locked
io.emit: PUBLISH incident_locked
incident_locked
{id, A, ttl}
200 {lockedBy, ttl}


Browser A
api pod 1
MongoDB
Redis
api pod 2
Browser B
(on pod 2)
PATCH .../:id
validate id, status and version
GET lock:incident:id
A (no lock, or another owner: 409)
updateOne({_id, version: v, status ∈ {open, escalated}},
{$set: status, closedBy; $inc: version})
matchedCount
alt [matchedCount = 1]
par [REST response]
EVAL RELEASE_LOCK_SCRIPT (A)
1: released
incident_unlocked
io.emit: PUBLISH incident_unlocked
incident_unlocked
incident_unlocked
200
[Change Stream: every pod has its own cursor]
change event
io.local.emit:
no Pub/Sub, no duplicates
change event
incident_update
incident_update
[matchedCount = 0]
findOne({_id})
current status, version
409
version changed (retry) or transition not allowed



The leader emits no event itself: every pod learns of the escalation from its own Change Stream.